

## Dataset Loading

The Iris dataset is loaded from the `iris.csv` file.

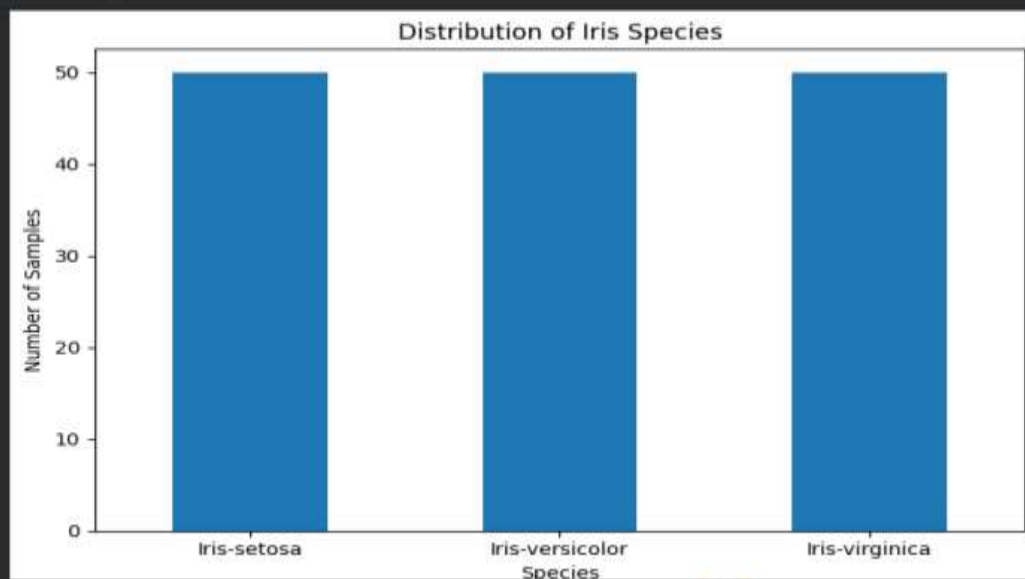
The dataset contains four numerical features and one categorical target variable representing the species of the flower.

```
data = pd.read_csv("iris.csv")  
print("Dataset Shape:", data.shape)  
display(data.head())
```

Dataset Shape: (150, 5)

|   | SepalLengthCm | SepalWidthCm | PetalLengthCm | PetalWidthCm | Species     |
|---|---------------|--------------|---------------|--------------|-------------|
| 0 | 5.1           | 3.5          | 1.4           | 0.2          | Iris-setosa |
| 1 | 4.9           | 3.0          | 1.4           | 0.2          | Iris-setosa |
| 2 | 4.7           | 3.2          | 1.3           | 0.2          | Iris-setosa |
| 3 | 4.6           | 3.1          | 1.5           | 0.2          | Iris-setosa |
| 4 | 5.0           | 3.6          | 1.4           | 0.2          | Iris-setosa |

```
data["Species"].value_counts().plot(  
    kind="bar"  
)  
  
plt.title("Distribution of Iris Species")  
plt.xlabel("Species")  
plt.ylabel("Number of Samples")  
plt.xticks(rotation=0)  
  
plt.tight_layout()  
plt.show()
```



## Loss Visualization

```
plt.figure(figsize=(8, 5))

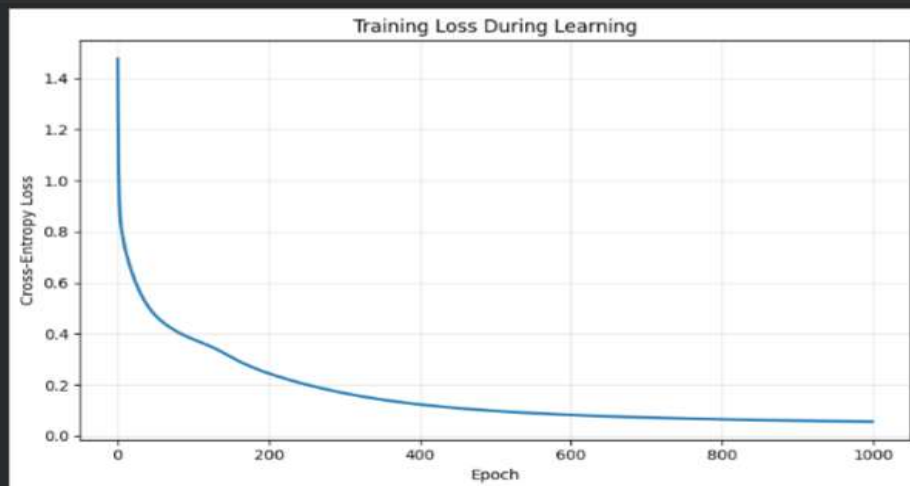
plt.plot(
    loss_history,
    linewidth=2
)

plt.title("Training Loss During Learning")
plt.xlabel("Epoch")
plt.ylabel("Cross-Entropy Loss")

plt.grid(True, alpha=0.3)

plt.tight_layout()

plt.show()
```



```
plt.show()
```

